Become an AI ML Expert with Phitron

Mathematics for Machine Learning

- Linear Equations & Linear Systems
- Gradient Descent
- Scalars & Vectors
- Probability Fundamentals
- Matrices & Matrix Multiplication
- Random Variables & Distributions
- Derivatives
- Derivatives of Different Linear Functions
- Discrete & Continuous Distributions
- Derivatives of Complex Functions
- Normal Distribution & Probability

Introduction to Python & Machine Learning

- Python Variables & Data Types
- List Basics & List Methods
- Operators in Python
- Lists as Stack & Queue
- Operator Precedence & Associativity
- List Comprehensions with Numbers & Strings
- For Loop
- While Loop
- Conditionals in Loops
- Break & Continue Statements
- Tuples & Sets in Python
- String Basics
- String Indexing & Slicing
- String Methods
- Dictionary Methods
- Dictionary Comprehensions
- Splitting, Joining & Formatting Strings
- Introduction to Machine Learning
- How a Machine Learning Model Works

AI Programming with Python

- Variables, Conditional Statements, Loops, Lists & Dictionaries Overview
- Introduction to OOP
- Classes & Objects
- Functions & Functional Programming
- Class & Object Theory
- Iterators & Generators
- Lambda Functions
- Map, Filter & Reduce Functions
- Class & Object Implementation
- File Handling
- Reading & Writing Files
- File Cursor
- Inheritance
- Polymorphism

- Practice Problems Based on File Handling
- Encapsulation
- Abstraction
- Exceptions
- Try, Except, Else & Finally
- **Project:** Movie Script Generator

Data Processing & Visualization (ML Toolbox)

NumPy

- NumPy Fundamentals
- `ndarray`
- Array Attributes & Data Types
- Array Creation from Existing Data
- Array Creation from Scratch
- Range-Based Arrays
- Random Arrays
- Array Manipulation
- Reshaping & Transformations
- Broadcasting
- Sorting, Searching & Counting
- Statistical Analysis
- Linear Algebra Basics

Pandas

- Pandas Series
- Pandas DataFrame
- File Handling
    - CSV
    - Excel
    - Other Data Formats
- Data Access
    - Indexing
    - `loc`
    - `iloc`
    - Filtering
- Indexing & Slicing
- Copying Data
- Advanced Iteration
- Modifying Data
    - Adding Rows
    - Dropping Rows
    - Adding Columns
    - Dropping Columns
    - Assigning Values
- Handling Duplicates
- Handling Missing Data
- DateTime Operations
- Arithmetic, Mathematical & Logical Operations
- Apply Functions
- Aggregation
- GroupBy Analysis

Matplotlib & Visualization

- Line Plot
- Scatter Plot
- Histogram
- Bar Plot
- Pie Chart
- Data Distribution Analysis
    - Percentiles
    - Histogram
    - Boxplot

Machine Learning Fundamentals

- Multiple Regression
- Polynomial Regression
- Mean, Median & Mode
- Variance & Standard Deviation
- End-to-End Regression Project
- Model Deployment
- Logistic Regression Classification
- Feature Relationship Analysis
    - Scatter Plot
    - Covariance
    - Correlation
- K-Nearest Neighbour (KNN)
- Exploratory Data Analysis (EDA)
- Data Cleaning
- Naive Bayes Classification
- Feature Engineering
- ML-Ready Data Preprocessing
- **Project:** Build a Machine Learning Model from Scratch
- **Projects:** Implement Machine Learning Algorithms

Advanced Machine Learning

Supervised Learning

- Decision Tree Classification
- Support Vector Machine (SVM)
- Rule-Based Learning Intuition
- Margin & Hyperplane
- Kernel Intuition
- Entropy
- Information Gain
- Supervised Model Performance Comparison
- Model Training
- Accuracy Evaluation
- Overfitting Control
- Tree Depth Tuning

Ensemble Learning

- Random Forest

- Decision Tree vs Random Forest
- Ensemble Learning Concepts

Unsupervised Learning

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN Density-Based Clustering
- Cluster Validation
- Cluster Visualization Techniques

Projects

- Student Performance Indicator
- Credit Card Fraud Detection
- Phishing Classifier

Deep Learning

Fundamentals

- Deep Learning Fundamentals
- Real-World Applications
- Machine Learning vs Deep Learning
- Neural Network Types
- Perceptron Concept
- Training & Loss Functions
- Limitations of Linear Models

PyTorch

- PyTorch Tensors
- Autograd
- Computational Graph
- Data Handling in PyTorch
- Training Pipeline
- Building Neural Networks in PyTorch

Neural Networks

- Multi-Layer Perceptron (MLP)
- Forward Propagation
- Backpropagation
- Backpropagation Mathematics
- ANN Classification
- ANN Regression

Optimization

- Momentum
- AdaGrad
- RMSProp
- Adam
- Weight Initialization

- Batch Normalization
- Normalization & Stability

Training Challenges

- Vanishing Gradients
- Exploding Gradients
- Feature Scaling
- Early Stopping
- Overfitting

Regularization

- L1 Regularization
- L2 Regularization
- Weight Decay
- Dropout

Projects

- Handwritten Digit Recognition (MNIST)
- Customer Churn Prediction
- House Price Prediction

CNN & RNN

Activation Functions

- Sigmoid
- Tanh
- ReLU
- Leaky ReLU
- ELU
- SELU

RNN

- Introduction to RNN
- Forward Propagation
- Types of RNN
- Advanced RNN
- Bidirectional RNN
- LSTM
- LSTM Architecture
- GRU
- Hyperparameter Tuning

CNN

- CNN Foundations
- CNN Architecture & Mathematics
- CNN in PyTorch
- CNN Architectures
- Data Augmentation

- Pre-trained CNN Models

Transformers

- Transformers & Modern Deep Learning
- Self-Attention
- Scaled Dot-Product Attention
- Transformer Encoder
- Transformer Decoder
- Transformer Inference

Projects

- Hate Speech Detection using BERT / Bangla BERT / mBERT
- Skin Lesion Classification
- Sentiment Analysis of Tweets
- Text Generation

Track 01: Project Track

Natural Language Processing (NLP)

- NLP Roadmap
- NLP Use Cases
- Core NLP Terminologies
- One-Hot Encoding
- NLP Pipeline
- Bag of Words (BoW)
- N-Grams
- Tokenization
- TF-IDF Representation
- Text Preprocessing
    - Stemming
    - Lemmatization
    - Stopword Removal
- Word Embeddings
- Word2Vec
    - CBOW
    - Skip-Gram
- POS Tagging
- Named Entity Recognition (NER)
- Word2Vec Implementation with Gensim
- Text Vectorization

NLP Projects

- Spam Detection
- Sentiment Analysis
- Text Generation

Large Language Models (LLMs)

- Evolution of NLP → Deep Learning → LLMs
- Introduction to Transformers
- LLM Roadmap
- Real-World LLM Use Cases
- Self-Attention
- Multi-Head Attention
- NLP Data Processing
- Sequence Data
- Positional Encoding
- Layer Normalization
- RNN Fundamentals
- Transformer Encoder–Decoder Architecture
- LSTM
- GRU
- Bidirectional RNN
- Transformer Language Models
- Text Generation
- Seq2Seq Encoder–Decoder Architecture
- Attention Mechanism
- NLP & LLM Applications
- LLM Deployment

Generative AI

Fundamentals

- Generative AI Ecosystem
- Generative AI Model Types
- Industry Use Cases
- Generative AI Career Roadmap
- LLM Fundamentals
    - Tokens
    - Embeddings
    - Transformers

LangChain

- LangChain Fundamentals
- LangChain Architecture
- Prompts
- Chains
- Runnables
- Output Parsers

RAG

- Documents
- Embeddings
- Vector Stores
- Retrievers
- Retrieval-Augmented Generation (RAG) Pipelines

LLM Workflows

- Prompting
- Fine-Tuning
- LLM Limitations
- Prompt Engineering
- Vector Databases
- LLMOps

AI Application Development

- Google Gemini Models & API with Python
- PDF Chat
- Document Question Answering
- Data Integration
- Text Generation
- Chatbots
- Vision Tasks
- Tools
- Agents
- Autonomous AI Workflows
- Local LLMs with Ollama
- Offline AI Applications

Deployment

- Streamlit
- Gradio
- LLM Application Scaling

Projects

- End-to-End Generative AI Projects

Track 02: Thesis Track

Part I — Topic Formulation & Scholarly Grounding

- Research vs Project vs Replication
- Types of Research Contributions
- Aligning Research Questions with Measurable Outcomes
- Research Ethics
- Academic Integrity
- Feasibility Analysis
    - Data
    - Compute
    - Tools
    - Approvals
- Topic Identification from Real-World Problems
- Identifying Research Gaps
- Risk Assessment
- Contingency Planning
- Scope Definition
- Systematic Literature Review
- Paper Selection
- Problem Statement

    - Context
    - Research Gap
    - Significance
- Literature Synthesis
- Positioning Against Prior Work
- Research Questions
- Research Objectives
- Research Design Blueprint
- Methodology
- Evaluation Plan

Part II — Data Strategy, Method Selection & Experimental Design

- Dataset Selection Criteria
    - Relevance
    - Bias
    - Licensing
    - Documentation
- Exploratory Data Analysis (EDA)
- Data Auditing
- Dataset Tracking
    - Source
    - Size
    - Features
    - Labels
    - Limitations
- Missing Data
- Label Noise
- Class Imbalance
- Ethical Data Use
    - Privacy
    - Consent
    - Legal Compliance
- Feature Checks
- Data Leakage Detection
- Data Diagnostics
- Data Collection Methods
- Sampling Strategy
- Defining Variables
- Defining Labels
- Evaluation Metrics
- Instrument Design
    - Surveys
    - Interviews
- Pilot Testing
- Baseline Models
- Benchmarking Strategy
- Data Management
    - Anonymization
    - Storage
    - Access Control
- Model Selection Rationale
- Experimental Design
- Reproducibility Plan
    - Train / Validation / Test Split
    - Hyperparameters

    - Version Control

Part III — Implementation, Optimization & Evidence-Based Reporting

- Reproducible Research Pipeline
- Repository Setup
- Data Preprocessing Pipelines
- Environment Management
- Baseline Implementation
- Benchmarking
- Sanity Checks
- Error Analysis
- Iterative Model Improvement
- Ablation Studies
- Robustness Evaluation
- Experiment Tracking
- Configuration Control
- Model Implementation
- Training Strategy
- Optimization Strategy
- Monitoring Overfitting
- Generalization
- Interpretability
- Fairness
- Ethical Considerations
- Results Presentation
    - Tables
    - Figures
    - Metrics
- Discussion
- Limitations
- Research Implications
- Paper Writing
- Publication Preparation
- Peer-Review Response